Graphs as Construction Histories: A Mixed-Radix Bijection via Matula Trees

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Abstract

We present a bijection between finite unlabeled graphs and integers based on vertex-by-vertex construction histories encoded in a mixed-radix system. The canonical representative is defined as the minimum mixed-radix integer over all labelings, yielding a global ordering that emphasizes build complexity rather than adjacency sparsity. The construction integer is itself a Matula number, providing an explicit rooted-tree interpretation. We compare this construction bijection with canonical adjacency, product invariants, and a block-cut "red-node" encoding, and we outline qualitative stress tests that distinguish these orderings on small graphs.

1 Introduction

Matula's bijection between rooted trees and integers gives a concrete way to read arithmetic structure as combinatorial structure. Extending this idea from rooted trees to general graphs raises two constraints: a canonical choice over labelings and a representation with interpretable structure. We propose a construction-based encoding that treats graphs as histories of incremental vertex additions.

Contributions.

1. A mixed-radix encoding of construction sequences that yields a bijection between graphs and integers once canonicalized by minimum integer.

2. A Matula-tree interpretation of each construction integer, yielding a structural diagram for the construction grammar.

3. A comparison framework that contrasts this ordering with canonical adjacency, product invariants, and block-cut (red-node) encodings.

2 Preliminaries

2.1 Graphs and labelings

All graphs are finite, undirected, and simple. For a graph on n vertices, consider all $n!$ labelings; canonicalization requires selecting a unique representative among these labelings.

2.2 Matula numbers

Matula's bijection maps rooted trees to integers via prime factorization [3, 1]. We assume the standard definitions; this paper uses the fact that any nonnegative integer corresponds to a unique rooted tree.

3 Construction sequences and mixed-radix encoding

Definition 1 (Construction sequence)**.** *Fix a labeling of a graph with vertices ordered as $v_0, v_1, \ldots, v_{n-1}$. For each $k \geq 1$, let $c_k \in \{0, 1, \ldots, 2^k - 1\}$ be the bitmask indicating which of $v_0, \ldots, v_{k-1}$ are adjacent to v_k. The sequence $[c_1, \ldots, c_{n-1}]$ is the* construction sequence *for the labeling.*

Definition 2 (Mixed-radix encoding)**.** *The construction integer is*

$$\mathrm{C}(G, \pi) = \sum_{k=1}^{n-1} c_k \prod_{j=1}^{k-1} 2^j,$$

where π is the chosen labeling order and the empty product is 1.

Example. For $n = 4$, the weights are $1, 2, 8$, so

$$\mathrm{C}(G, \pi) = c_1 + 2c_2 + 8c_3.$$

Definition 3 (Canonical construction integer)**.** *The canonical construction integer of an unlabeled graph G is*

$$\mathrm{C}(G) = \min_{\pi} \mathrm{C}(G, \pi),$$

where π ranges over all labelings.

Proposition 1. *For fixed n, the mapping from construction sequences to integers is a bijection.*

Theorem 1 (Canonical construction bijection (fixed n))**.** *For each unlabeled graph G on n vertices, the canonical construction integer $\mathrm{C}(G)$ is well-defined and unique. The mapping that sends G to $\mathrm{C}(G)$ is injective on unlabeled graphs with n vertices, and the mixed-radix encoding provides a decoding from integers to construction sequences for each fixed n.*

Proof sketch. For fixed n, the mixed-radix encoding is a bijection between sequences $[c_1, \ldots, c_{n-1}]$ and integers in $[0, \prod_{k=1}^{n-1} 2^k)$, so decoding is unique. For any unlabeled graph G, the set of labelings is finite, so the minimum construction integer exists. If two graphs had the same canonical integer, then they would share a construction sequence under some labeling, which determines the adjacency pattern and hence the isomorphism class.

Remark (global bijection). To make the encoding injective across all sizes, one can prefix the size (or use a STOP symbol in a unified mixed-radix grammar). This paper focuses on the fixed n encoding and uses size as an external parameter when ordering graphs across different sizes.

4 Matula trees for construction integers

Because each construction integer is a nonnegative integer, it corresponds to a unique rooted tree under the Matula bijection. This yields a compositional tree diagram for the construction grammar of the graph (see Fig. 1).

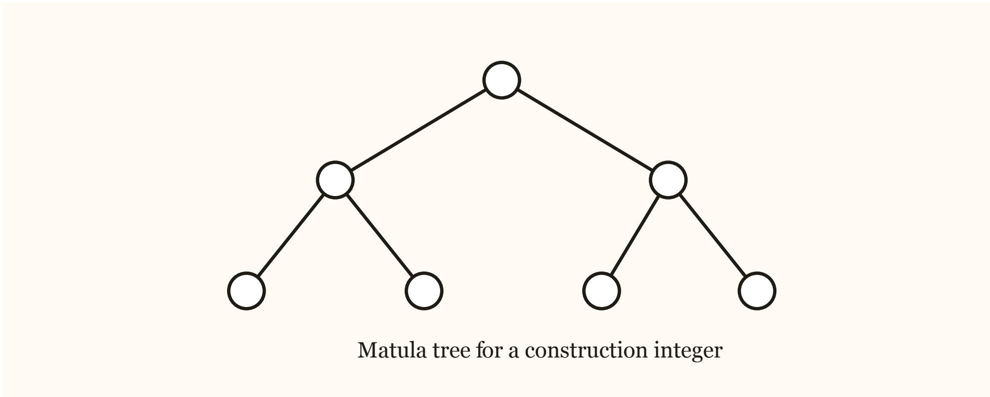

Figure 1: A Matula tree diagram for a construction integer.

5 Comparisons and stress tests

We contrast the construction bijection with three alternatives: canonical adjacency encoding, product invariants over all labelings, and a block-cut red-node encoding. We emphasize examples where the orderings diverge (e.g., star vs path in Fig. 2, cycle vs matching in Fig. 3, and barbell vs two triangles in Fig. 4).

Red-node encoding (sketch). The red-node view decomposes a graph into its block-cut tree and attaches a construction encoding to each 2-connected block. The red level marks the boundary between topology (how blocks connect) and construction (what each block is), producing a tree diagram that preserves structural semantics but is not intrinsically dense.

Small $n = 5$ example list. Ordering the unlabeled graphs by canonical adjacency and listing the resulting construction integers for $n = 5$ gives:

$0, 1, 3, 11, 75, 7, 12, 13, 15, 76, 77, 79, 30, 31, 86, 87, 94, 95, 222, 223, 63, 116, 117, 119, 127, 235, 236, 237, 239, 254, 255, 50$

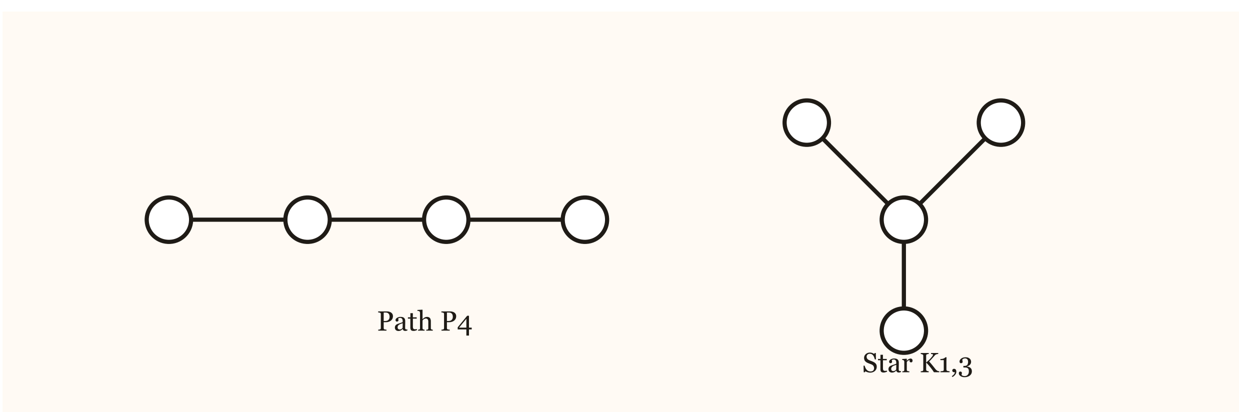

Figure 2: Path vs star: same size, different construction profiles.

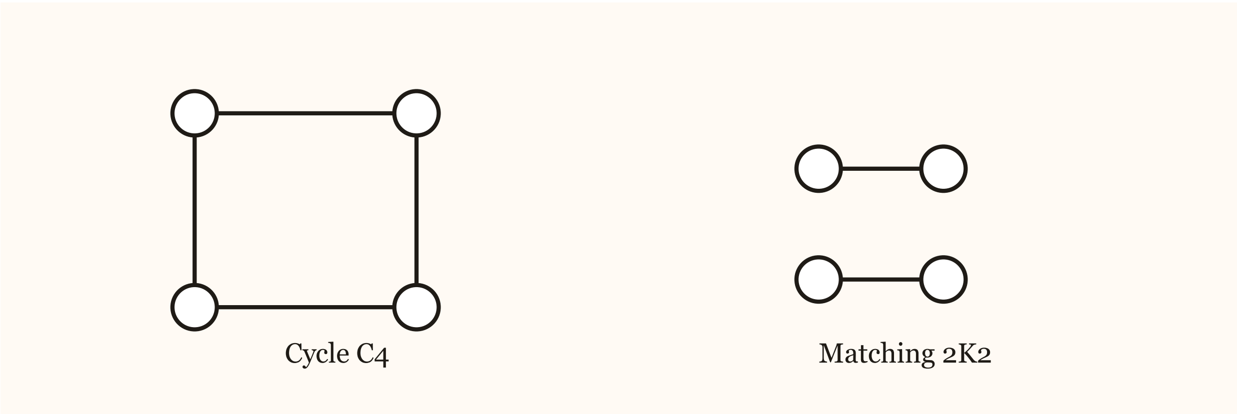

Figure 3: Cycle vs matching: same n and m, different symmetry profiles.

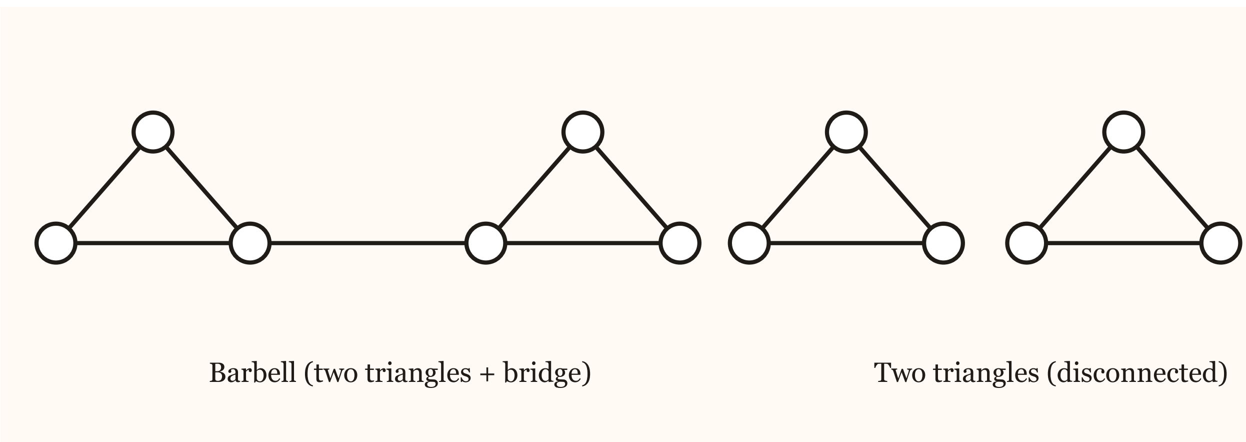

Figure 4: Barbell vs two disconnected triangles: same blocks, different topology.

6 Complexity and canonicalization

Canonical construction requires considering all $n!$ labelings in the worst case, matching the known hardness of graph canonicalization. The contribution is structural, not algorithmic: the ordering makes constructional complexity visible rather than hidden. For practical canonical labeling algorithms, see the nauty/traces line of work [4].

Naive canonicalization algorithm. Given a graph on n vertices, enumerate all labelings, compute the construction integer for each labeling, and return the minimum. This brute-force procedure is feasible only for small n, but it makes the canonicalization requirement explicit.

7 Related work

Canonical adjacency encodings of unlabeled graphs appear in the literature as "mincode" or graph-code numbers: one minimizes the lower-triangular adjacency bitstring over all labelings and interprets it as an integer [2, 5]. OEIS sequences A076184 and A382754 enumerate these canonical adjacency codes, and A382757 maps the corresponding graph list to those codes. Our work differs in the encoding semantics: the integer is derived from a construction history (mixed-radix sequence) and then decoded as a Matula tree, which supplies a compositional interpretation beyond canonical adjacency.

For trees, the classical Prüfer code gives a bijection between labeled trees and integer sequences [6]. The construction sequences here can be seen as a higher-level analog that applies to general graphs.

8 Discussion and open questions

We outline possible correlations between construction complexity and classical graph invariants, and ask whether block-cut decomposition can be combined with construction encoding to reduce collisions while preserving structure.

References

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